

L'ACADIE, QC, Canada

WMO: 713720

Lat: 45.2939N

Lon: 73.3497W

Elev: 44

StdP: 100.80

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.1	-21.5	-28.9	0.3	-23.5	-26.2	0.3	-21.1	13.0	0.2	11.4	0.2	2.8	270	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	30.1	21.7	28.6	21.0	27.2	20.3	23.6	27.8	22.6	26.4	21.7	25.3	4.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	17.0	26.1	21.3	16.0	24.9	20.3	15.1	23.9	70.9	28.1	66.7	26.5	63.3	25.4	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	10.2	8.7	7.5	DB	-27.5	32.9	3.7	1.5	-30.2	33.9	-32.3	34.8	-34.4	35.6	-37.1	36.7
(5)				WB	-27.8	25.6	3.6	1.3	-30.4	26.5	-32.5	27.2	-34.5	27.9	-37.1	28.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.0	-9.1	-7.6	-2.0	6.0	13.8	18.8	21.0	19.8	15.8	9.0	2.3	-4.6
	DBStd	11.68	7.34	6.30	6.08	4.62	4.32	3.66	2.76	2.91	3.98	4.50	5.09	6.32
	HDD10.0	2372	593	494	372	136	13	0	0	0	3	74	234	453
	HDD18.3	4369	851	727	629	370	152	37	7	19	96	290	481	711
	CDD10.0	1280	0	0	1	16	130	265	341	303	177	44	4	0
	CDD18.3	237	0	0	0	1	11	52	89	63	19	1	0	0
	CDH23.3	1774	0	0	0	7	128	414	650	426	145	4	0	0
	CDH26.7	413	0	0	0	1	31	114	155	84	29	0	0	0
Wind	WSAvg	3.2	3.8	3.6	3.5	3.6	3.3	3.0	2.6	2.4	2.6	3.1	3.5	3.4
Precipitation	PrecAvg	1031	73	61	68	84	89	97	106	96	90	97	78	88
	PrecMax	1224	115	100	117	152	180	148	175	163	201	176	156	167
	PrecMin	872	24	27	21	14	31	51	49	41	33	31	20	33
	PrecStd	105	28	21	23	37	36	30	28	33	36	41	29	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.7	7.3	15.3	23.7	30.1	32.0	32.3	30.8	29.9	23.6	17.3	10.9
		MCWB	7.9	5.4	10.1	14.8	18.8	22.5	23.6	23.2	22.3	17.4	14.1	9.1
	2%	DB	5.1	4.8	10.8	19.6	26.8	29.7	29.9	28.8	26.8	20.7	14.4	7.6
		MCWB	3.7	3.1	7.3	12.3	17.0	20.9	22.3	21.8	20.6	15.9	12.1	6.3
	5%	DB	3.1	3.1	7.9	16.4	24.1	27.7	28.3	27.3	24.8	18.3	11.7	5.4
		MCWB	2.1	1.6	5.4	10.5	15.4	19.7	21.4	20.9	19.3	14.3	9.5	4.2
	10%	DB	1.4	1.7	5.7	13.6	21.7	25.6	26.9	25.8	22.8	16.0	9.7	3.3
		MCWB	0.4	0.6	3.5	8.6	14.9	18.6	20.7	20.1	18.2	12.9	7.7	2.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.1	5.7	11.8	17.1	21.0	24.3	25.6	24.6	23.2	18.9	14.9	9.7
		MCDB	8.7	7.1	14.4	21.1	27.1	29.6	30.6	29.0	27.3	22.0	16.4	10.8
	2%	WB	4.0	3.3	7.8	13.8	19.2	22.7	23.9	23.1	21.6	16.7	12.2	6.6
		MCDB	4.8	4.6	10.2	18.5	24.0	27.2	28.0	27.0	25.3	19.9	14.1	7.5
	5%	WB	2.1	1.8	5.6	11.2	17.6	21.3	22.7	22.0	20.2	14.7	9.9	4.3
		MCDB	2.9	2.9	7.5	14.9	21.9	25.4	26.3	25.5	23.7	17.7	11.4	5.3
	10%	WB	0.4	0.6	3.7	9.4	15.9	20.1	21.8	21.1	19.0	13.2	7.8	2.4
		MCDB	1.3	1.6	5.5	13.1	20.1	24.0	25.4	24.3	22.1	15.5	9.6	3.2
Mean Daily Temperature Range	5% DB	MDBR	8.6	9.0	8.5	9.5	10.7	9.9	9.9	10.3	10.6	8.7	7.1	7.3
		MCDBR	8.3	8.5	10.4	14.8	14.5	12.5	11.6	11.9	12.6	11.5	10.2	7.6
		MCWBR	8.0	7.6	7.7	8.8	7.1	5.9	5.9	6.2	6.9	7.2	8.4	7.1
	5% WB	MCDBR	8.1	8.3	10.1	12.9	11.6	10.1	9.8	10.2	10.9	10.3	9.6	7.6
		MCWBR	8.0	7.7	8.1	8.6	7.1	5.9	5.8	5.9	6.6	7.2	8.5	7.2
Clear-Sky Solar Irradiance	taub	0.264	0.291	0.326	0.363	0.390	0.440	0.436	0.418	0.376	0.335	0.309	0.297	
	taud	2.364	2.278	2.335	2.356	2.338	2.204	2.222	2.264	2.371	2.491	2.484	2.318	
	Ebn at Noon	857	890	905	896	879	832	830	832	840	835	797	760	
	Edh at Noon	74	98	108	116	122	140	136	126	104	80	67	71	
All-Sky Solar Radiation	RadAvg	1.45	2.37	3.58	4.34	5.10	5.59	5.63	5.00	3.98	2.30	1.41	1.01	
	RadStd	0.16	0.24	0.28	0.52	0.50	0.44	0.42	0.23	0.33	0.21	0.18	0.09	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page